

AI virtual-camera tracking that finds, identifies and holds the partner terminal's beacon, in software

[Proposed solution](#) · [Detailed explanation](#) · [How it addresses the problem](#) · [Innovation and uniqueness](#)

PROBLEM EXISTING

- Micro-radian aim**
A hair-thin laser must stay on target while vehicles, drones and masts move. (Kaymak 2018)
- Wrong-target lock**
Stars, glints and lamps look like the beacon; brightness-only trackers lock the wrong light.
- Delay & dropouts**
≈40 ms command lag and blink gaps drop the beam, forcing slow manual re-alignment.

PROPOSED SOLUTION

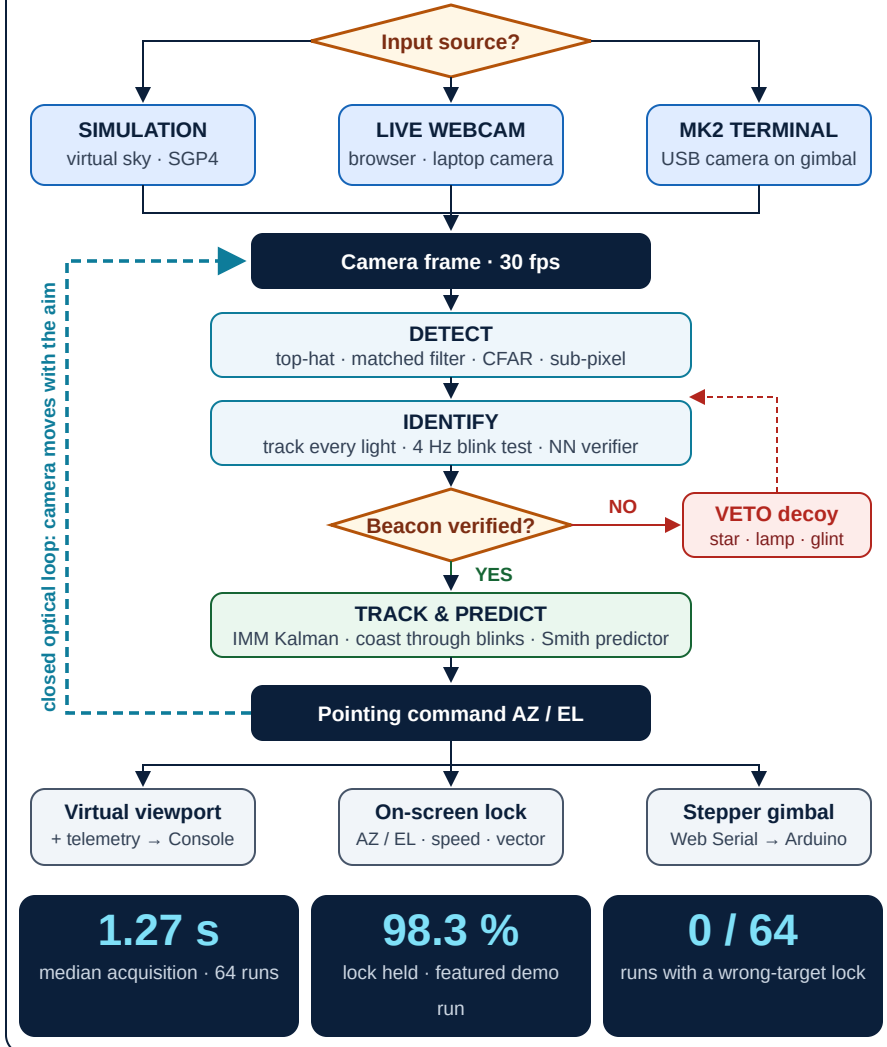
We propose **ZeroDrift**: software that finds the partner's **4 Hz beacon** in a camera image, **proves** it is the right light and **steers** the terminal to hold it centred: PS26169 solved without special hardware.

- Physics-true virtual camera**
SGP4 orbits, turbulence, decoys: **tested before hardware**
- Detect**
top-hat · matched filter · CFAR · **sub-pixel**
- Identify**
4 Hz blink test + neural verifier: **0 / 64 wrong locks**
- Track & coast**
IMM Kalman holds through blinks: **98.3 % held**
- Control**
Smith predictor cancels the command delay: **198 μ rad** median error, lock in **1.27 s**

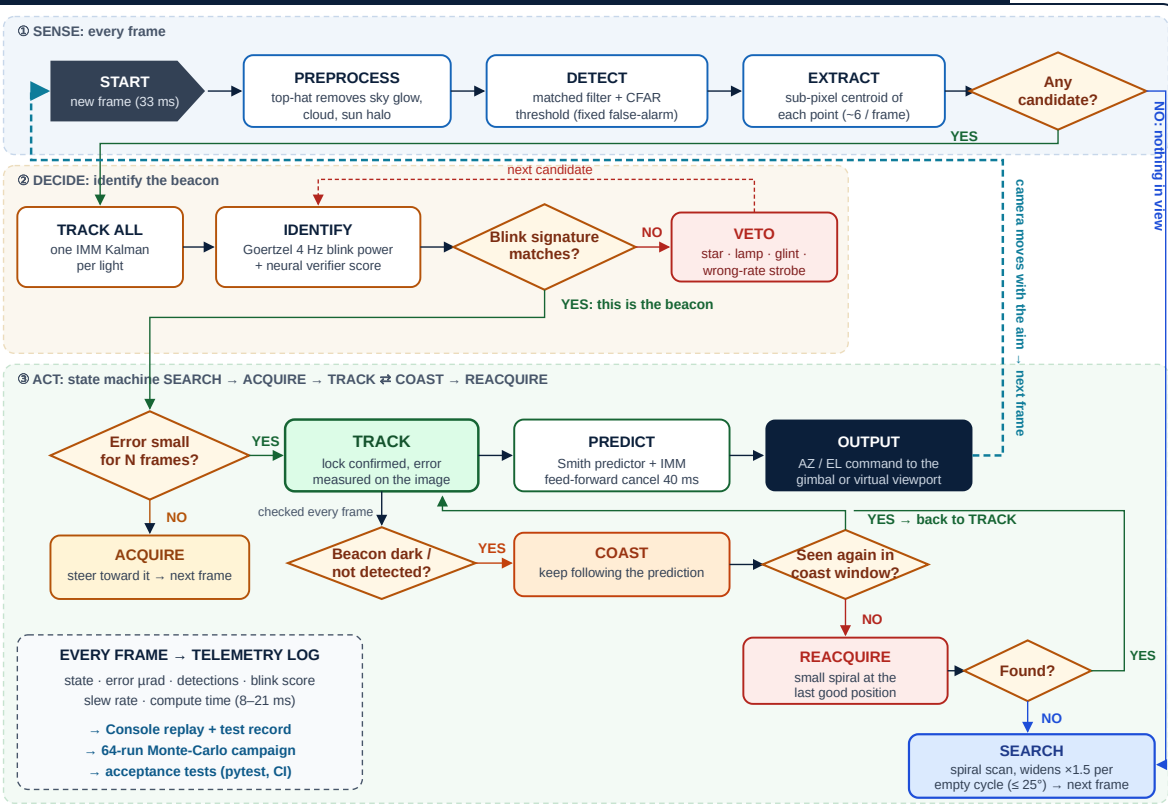
UVP (UNIQUE VALUE PROPOSITION)

- Identity, not brightness**
Locks only on the blink signature
- Delay-cancelled loop**
Error measured on the image itself
- One engine, 3 modes**
Simulation · webcam · MK2 terminal
- Measured, not claimed**
Open source · 250 passing tests

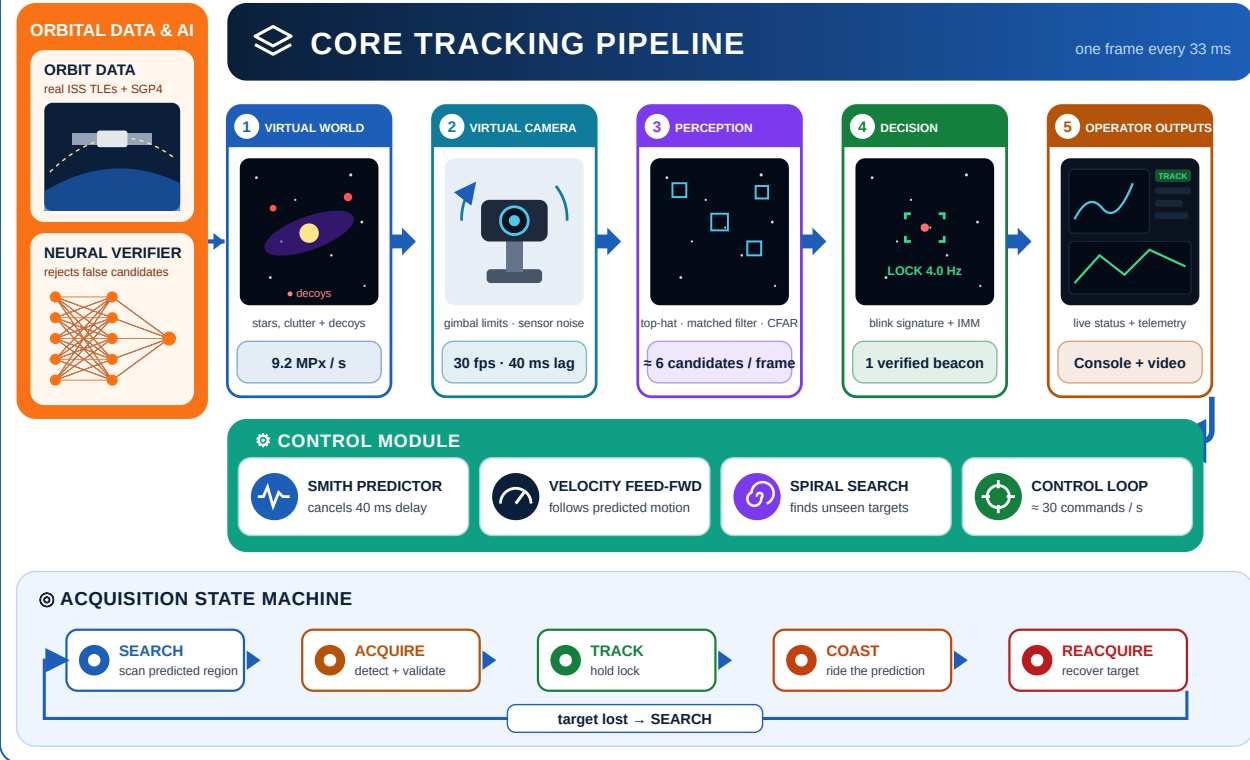
ARCHITECTURE



SOFTWARE FLOWCHART · src/fsoc_pat (pipeline · detection · tracking · control)



SYSTEM ARCHITECTURE · WHAT HAPPENS TO EVERY FRAME



TECH STACK · every item used in the code

ENGINE

Py Python 3.11

Np NumPy

CV OpenCV

Or SGP4 · ISS TLEs

Ym PyYAML scenes

AI / LOGIC

Cf CFAR detector

Km IMM Kalman

HZ Goertzel 4 Hz

NN Neural verifier

Sp Smith predictor

APPS & WEB

Qt PySide6 GUI

Pg PyQtGraph

JS JavaScript

3D three.js twin

Rt WebRTC camera

Se Web Serial

HARDWARE

Ar Arduino C++

As AccelStepper

St A4988 + NEMA17

I2 AS5600 · I²C

QUALITY & DEPLOY

Pt pytest · 250 pass

CI GitHub Actions

Ex PyInstaller app

Nf Netlify site

PROOF > PROMISE · WORKING PROTOTYPE

